

Violence Against Civilians and Leader Favorability in Mali

Multilevel models of commune-level ACLED events and a three-wave household survey

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1 Question

Is violence against civilians in a respondent's commune associated with how favorably they view the national leader, comparing people in the same region and survey wave? Violence is the number of ACLED events of violence against civilians in the commune during the same window as each wave's survey fieldwork. Favorability is a yes/no item, and most respondents answer yes, so the association is expected to be modest and the analysis is built to estimate it carefully rather than to find it.

2 Data and design

The analysis file has 5,892 respondents (67 dropped for missing data) in 174 communes. Communes surveyed in one, two, and three waves: 14, 83, 77. Each wave is an independent two-stage sample within regions: communes first, then villages within communes.

Table 1: Sample by wave, with the design-weighted percent favorable (design-based SE) and the median number of events per respondent's commune.

Wave	Respondents	Communes	Villages	% favorable	SE	Median events
1	1964	137	411	76.7	1.6	1
2	1967	137	411	78.7	1.7	1
3	1961	137	411	73.5	1.5	2

3 Models

3.1 Violence measure

For commune c in wave w , with E_{cw} events,

$$V_{cw} = \log(1 + E_{cw}).$$

Zero events gives $V = 0$. Adding 1 to V multiplies $(1 + E)$ by $e \approx 2.72$; adding $\log 2 \approx 0.69$ doubles it. Because these steps are hard to picture, results are also shown as predicted favorability at 0, 1, 5, and 20 events. Only 10% of the variation in V_{cw} is within communes across waves; the rest is between communes.

3.2 Equations

Respondent i lives in commune c and was interviewed in wave w ; $y_{icw} = 1$ if they view the leader favorably. All models are logistic multilevel models with communes as the grouping level; M3b adds region-wave cells and M4 villages.

M1. Empty model.

$$\text{logit Pr}(y_{icw} = 1) = \gamma_{00} + u_{0c}, \quad u_{0c} \sim N(0, \tau_{00})$$

γ_{00} is the log-odds of a favorable view in a typical commune and τ_{00} the variance of communes around it. A logistic model has no residual variance term, so the intraclass correlation uses the variance of the standard logistic distribution, $\pi^2/3 \approx 3.29$:

$$\text{ICC} = \frac{\tau_{00}}{\tau_{00} + \pi^2/3}$$

M2a. Violence, random intercept.

$$\text{logit Pr}(y_{icw} = 1) = \gamma_{00} + \gamma_{10}V_{cw} + u_{0c}$$

M2b. Violence, random intercept and random slope.

$$\text{logit Pr}(y_{icw} = 1) = \gamma_{00} + \gamma_{10}V_{cw} + u_{0c} + u_{1c}V_{cw}$$

$$\begin{pmatrix} u_{0c} \\ u_{1c} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_{00} & \tau_{01} \\ \tau_{01} & \tau_{11} \end{pmatrix}\right)$$

γ_{10} is the violence effect in a typical commune; τ_{11} is how much that effect varies across communes; τ_{01} is the covariance between a commune's baseline favorability and its violence effect.

Random slope test. $H_0 : \tau_{11} = 0$ against $H_A : \tau_{11} > 0$. The statistic is $\text{LRT} = -2(\ell_{\text{M2a}} - \ell_{\text{M2b}})$. Because a variance cannot be negative, the null value is on the boundary of the parameter space, and the p-value comes from a 50:50 mixture of χ_1^2 and χ_2^2 distributions. The decision rule was fixed in advance: keep the random slope only if $p < .05$ and M2b is not singular.

M3a. Region, wave, and covariates. The M2 winner plus region dummies (Bamako is the reference), wave dummies (wave 1 is the reference), and respondent covariates $\mathbf{x}_i$ (female, age per decade centered, ethnicity, education, urban):

$$\text{logit Pr}(y_{icw} = 1) = \gamma_{00} + \gamma_{10}V_{cw} + \sum_{r \neq \text{Bamako}} \alpha_r \text{Region}_r + \gamma_2 \text{Wave}_2 + \gamma_3 \text{Wave}_3 + \mathbf{x}'_i \boldsymbol{\beta} + u_{0c} [+u_{1c}V_{cw}]$$

The term in brackets is included in M3a, M3b, and M4 only if the random slope test keeps it.

M3b. Region-by-wave shocks. M3a plus a random intercept for each region-wave cell, $v_{rw} \sim N(0, \tau_{rw})$, independent of u_{0c} :

$$\text{logit Pr}(y_{icw} = 1) = \text{M3a} + v_{rw}$$

The region and wave dummies stay. v_{rw} captures shifts that hit one region in one wave (a local event, a security operation) and that separate region and wave dummies cannot absorb; with 24 cells, the cells with few respondents are pulled toward the region and wave main effects (partial pooling). Communes are nested in regions and crossed with waves, which `lme4` handles directly.

Region-by-wave test. $H_0 : \tau_{rw} = 0$. M3b adds one variance, so the boundary mixture is $0.5\chi_0^2 + 0.5\chi_1^2$ (half the usual χ_1^2 p-value). The rule was fixed in advance: M3b proceeds only if $p < .05$, its AIC is lower than M3a's, and it is not singular. The winner is called M3 below.

M4. Final model: M3 plus the survey design. The design enters the multilevel model directly (Pfeffermann et al., 1998; Gelman, 2007): strata are already the region dummies, first-stage units (communes) are already a random intercept, and M4 adds the second stage, villages within communes, $w_{vc} \sim N(0, \tau_v)$, and the centered log of the survey weight as a covariate:

$$\text{logit Pr}(y_{vcw} = 1) = \text{M3} + \delta \log \tilde{w}_i + w_{vc}$$

τ_v shows how much villages differ within communes. δ checks for informative sampling: if respondents with larger weights differ in favorability after the predictors, δ is clearly nonzero and estimates that ignore the weights may not represent the population. Sampling weights themselves cannot be passed to `glmer`: its `weights` argument means something else for binomial models.

Design-based check. For comparison, the same fixed effects are fit with `survey::svyglm`: weights rescaled within wave ($w_i^* = w_i n_w / \sum_{i \in w} w_i$), strata = regions within wave, first stage = communes within wave, second stage = villages, and Taylor-linearized standard errors at the first stage (Binder, 1983):

$$\widehat{\text{Var}}(\hat{\beta}) = \hat{\mathbf{J}}^{-1} \left[\sum_h \frac{n_h}{n_h - 1} \sum_{j=1}^{n_h} (\mathbf{z}_{hj} - \bar{\mathbf{z}}_h)(\mathbf{z}_{hj} - \bar{\mathbf{z}}_h)' \right] \hat{\mathbf{J}}^{-1}, \quad \mathbf{z}_{hj} = \sum_{i \in (h,j)} w_i^* \mathbf{x}_i (y_i - \hat{\mu}_i)$$

Its coefficient is population-averaged, smaller than the multilevel (typical-commune) coefficient by roughly $\sqrt{1 + 0.346 \tau_{00}}$ (Zeger, Liang & Albert, 1988), so the two are compared by z statistics, not raw coefficients.

Sensitivity: baseline level and change (M3-MW). Violence is split into a commune's usual level, $\bar{V}_c$ (the mean of V_{cw} over the waves in which it was surveyed), and its deviation in a given wave (Mundlak, 1978; Bell & Jones, 2015), with the same random part as M3:

$$\text{logit Pr}(y_{icw} = 1) = \gamma_{00} + \gamma_W(V_{cw} - \bar{V}_c) + \gamma_B \bar{V}_c + \dots + u_{0c}$$

γ_B compares communes with persistently more violence to those with less; γ_W asks whether a commune is less favorable in a wave when its violence is above its usual level. If the two are similar, the single coefficient is a fair summary.

Estimation. All multilevel models are fit by maximum likelihood with the Laplace approximation (`lme4::glmer`, `bobyqa` optimizer), without weights, on the same respondents, so their likelihoods, AIC, and BIC can be compared. The design-based check has a pseudo-likelihood and is never compared by AIC or likelihood ratio tests.

4 Results

4.1 M1: how much favorability differs between communes

Observations	5892
Dependent variable	fav
Type	Mixed effects generalized linear model

Family binomial
Link logit

AIC 6331.478

BIC 6344.841

Fixed Effects				
	Est.	S.E.	z val.	p
(Intercept)	1.219	0.061	19.874	0.000

Random Effects		
Group	Parameter	Var.
commune_id	(Intercept)	0.453

Grouping Variables		
Group	# groups	ICC
commune_id	174	0.121

The commune variance is $\hat{\tau}_{00} = 0.453$, so the ICC is 0.121: 12% of the variation in the latent propensity to view the leader favorably lies between communes. Respondents in the same commune are more alike than respondents in different communes, which is why every model that follows keeps a commune random intercept. In a typical commune, the predicted percent favorable is 77.2%.

4.2 M2: violence and the random slope test

	M2a: random intercept	M2b: random slope
Violence, log(1 + events)	-0.342 *** [-0.430, -0.255]	-0.345 *** [-0.432, -0.259]
N	5892	5892
N (commune_id)	174	174
AIC	6280.799	6284.774
BIC	6300.843	6318.181
R2 (fixed)	0.037	0.037
R2 (total)	0.118	0.118

*** p < 0.001; ** p < 0.01; * p < 0.05.

Table 2: Likelihood ratio test for the random violence slope (50:50 mixture of chi-square with 1 and 2 df).

Model	Log-likelihood	LRT	p (mixture)	Singular
M2a: random intercept	-3137.40			FALSE
M2b: + random slope	-3137.39	0.02	0.931	TRUE

The likelihood ratio statistic is 0.02 with mixture $p = 0.931$, and M2b is singular (at least one variance or correlation is estimated at its boundary). The random slope is not kept, so M3a, M3b, and M4 have a commune random intercept only. This should not be read as proof that violence affects every commune in the same way. A random slope for a commune-level exposure is identified only by changes in violence within the same commune across waves, and only 10% of the variation in violence is of that kind. The data have little information to estimate the slope variance.

4.3 M3: region, wave, and region-by-wave shocks

	M3a	M3b: + region-wave
Violence, log(1 + events)	-0.295 *** [-0.386, -0.205]	-0.300 *** [-0.391, -0.209]
Wave 2	0.283 *** [0.122, 0.443]	0.287 * [0.048, 0.526]
Wave 3	0.043 [-0.113, 0.199]	0.040 [-0.196, 0.275]
Female	0.238 *** [0.112, 0.364]	0.238 *** [0.112, 0.364]
Age (per decade, centered)	0.073 ** [0.021, 0.125]	0.072 ** [0.020, 0.124]
Urban	-0.273 * [-0.508, -0.038]	-0.289 * [-0.526, -0.052]
N	5892	5892
N (commune_id)	174	174
AIC	6231.315	6223.257
BIC	6384.986	6383.610
R2 (fixed)	0.072	0.072
R2 (total)	0.135	0.144
N (region_wave)		24

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table 3: Likelihood ratio test for the region-by-wave random intercept (50:50 mixture of chi-square with 0 and 1 df).

Model	Log-likelihood	AIC	BIC	LRT	p (mixture)	Singular
M3a	-3092.66	6231.31	6384.99			FALSE
M3b: + region-wave	-3087.63	6223.26	6383.61	10.06	< .001	FALSE

The region-by-wave variance is $\hat{\tau}_{rw} = 0.031$, with likelihood ratio statistic 10.06 and mixture $p < .001$; AIC is lower for M3b. M3b proceeds: favorability shifts by region and wave in ways the separate dummies do not capture, and those shifts are now kept out of the violence coefficient. The violence coefficient is -0.300 in M3b against -0.295 in M3a.

4.4 M4: final model with the survey design

Observations	5892
Dependent variable	fav
Type	Mixed effects generalized linear model
Family	binomial
Link	logit

AIC 6226.394

BIC 6400.109

Fixed Effects					
	Est.	2.5%	97.5%	z val.	p
(Intercept)	1.597	0.977	2.216	5.048	0.000
v_commune	-0.303	-0.395	-0.211	-6.463	0.000
regionGao	-0.408	-1.091	0.275	-1.169	0.242
regionKayes	0.136	-0.522	0.795	0.405	0.685
regionK-oulikoro	0.221	-0.428	0.869	0.666	0.505
regionMopti	-0.296	-0.965	0.373	-0.868	0.386
regionSégou	-0.150	-0.814	0.514	-0.443	0.657
region-Sikasso	-0.090	-0.739	0.559	-0.273	0.785
region-Tombouctou	0.060	-0.606	0.726	0.178	0.859
wave2	0.287	0.048	0.527	2.352	0.019
wave3	0.038	-0.198	0.274	0.314	0.754

Fixed Effects					
	Est.	2.5%	97.5%	z val.	p
female	0.240	0.114	0.366	3.719	0.000
age10	0.073	0.021	0.125	2.729	0.006
ethnicityFu-lani	-0.281	-0.469	-0.092	-2.915	0.004
ethnicitySonghai	0.109	-0.138	0.357	0.865	0.387
ethnicitySoninke	0.004	-0.235	0.244	0.033	0.974
ethnicityOther	0.191	-0.061	0.442	1.483	0.138
ethnicityDogon	0.098	-0.154	0.350	0.760	0.447
ethnicityTwareg	-0.336	-0.598	-0.074	-2.511	0.012
education-Primary	-0.059	-0.208	0.090	-0.779	0.436
education-Secondary+	-0.141	-0.297	0.015	-1.774	0.076
urban	-0.289	-0.527	-0.051	-2.378	0.017
log_wt_c	-0.001	-0.075	0.072	-0.034	0.973

Random Effects		
Group	Parameter	Var.
commune_id:village	(Intercept)	0.046
commune_id	(Intercept)	0.242
region_wave	(Intercept)	0.030

Grouping Variables		
Group	# groups	ICC
commune_id:village	1031	0.013
commune_id	174	0.067
region_wave	24	0.008

Violence. Holding region, wave, respondent characteristics, and the design terms constant, a one-unit increase in $\log(1 + E)$ changes the log-odds of a favorable view by -0.303 (95% CI -0.395 to -0.211) in a

typical commune, village, and region-wave. As an odds ratio, the odds of a favorable view are multiplied by 0.74 each time $(1 + E)$ is multiplied by 2.72, and by 0.81 each time $(1 + E)$ doubles. On the probability scale, the average marginal effect is -5.2 percentage points per unit (95% CI -6.7 to -3.6).

Region and wave. Region coefficients compare each region with Bamako, and wave coefficients compare waves 2 and 3 with wave 1, holding the other predictors constant. Positive values mean higher favorability than the reference.

Variance components. After the predictors, communes vary with $\hat{\tau}_{00} = 0.246$ in M3 (residual ICC 0.070, down from 0.121 in M1), and villages within communes with $\hat{\tau}_v = 0.046$.

Informative sampling. The log-weight coefficient is -0.001 (SE 0.037, $p = 0.973$). There is no clear sign that the weights carry information about favorability beyond the predictors, which supports the unweighted multilevel estimates.

4.5 Cost of the design terms and the design-based check

Table 4: Violence coefficient (log-odds) across models. M2a to M4 are conditional (typical commune); the design-based check is population-averaged, so compare z , not the coefficient.

Model	Estimate	SE	z
M2a	-0.342	0.045	-7.69
M3b	-0.300	0.047	-6.45
M4 (final)	-0.303	0.047	-6.46
Design-based (svyglm)	-0.254	0.053	-4.81

Adding the design terms changes the standard error of the violence coefficient by a factor of 1.01 and the coefficient by a factor of 1.01 (M4 against M3). The design-based check, which uses the weights directly, gives $z = -4.81$ against -6.46 for M4. The two lead to the same conclusion.

4.6 Predicted favorability

Table 5: M4: predicted percent with a favorable view if every respondent's commune had the given number of events, other characteristics as observed, typical commune and village (95% CI).

Events in commune	Predicted % favorable	95% CI
0	82.3	80.0 to 84.6
1	79.1	77.0 to 81.2
5	73.2	70.6 to 75.8
20	65.4	60.7 to 70.1

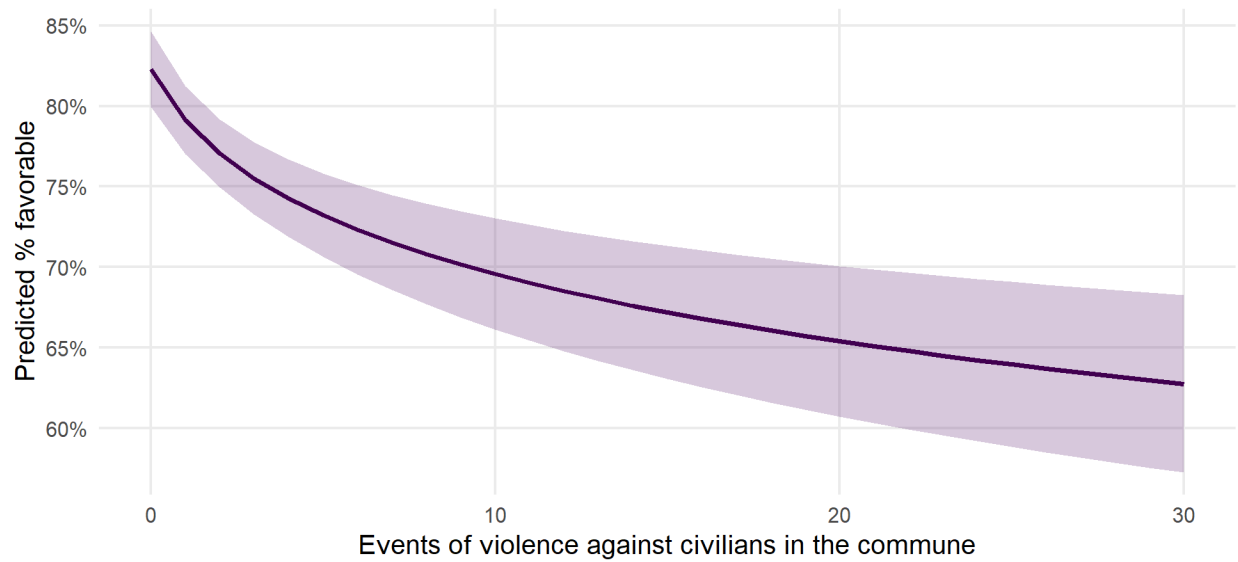


Figure 1: M4: predicted percent favorable across event counts (95% CI). The x-axis is the event count; the model is linear in $\log(1 + \text{events})$.

4.7 Sensitivity: usual level of violence versus change

	M3	M3-MW
Violence, $\log(1 + \text{events})$	-0.300 *** [-0.391, -0.209]	
Violence, within commune		-0.246 * [-0.439, -0.053]
Violence, between communes		-0.316 *** [-0.419, -0.212]
N	5892	5892
N (commune_id)	174	174
N (region_wave)	24	24
AIC	6223.257	6224.873
BIC	6383.610	6391.907
R2 (fixed)	0.072	0.073
R2 (total)	0.144	0.145

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

The between-commune coefficient is -0.316 and the within-commune coefficient -0.246 . The between part compares communes with persistently different violence; the within part uses only communes surveyed in more than one wave whose violence changed, so it is less precise.

5 Diagnostics

Table 6: Fit of the multilevel models (same respondents, maximum likelihood). ICC here is from `performance::icc()` and combines all random effects in the model (commune, and region-wave and village where included), so it differs from the commune-only ICC in the text. NA: not computed for a singular fit (M2b).

Model	AIC	BIC	ICC
M1	6331.48	6344.84	0.121
M2a	6280.80	6300.84	0.084
M2b	6284.77	6318.18	NA
M3a	6231.31	6384.99	0.069
M3b	6223.26	6383.61	0.078
M4	6226.39	6400.11	0.088

Table 7: Collinearity of the M4 predictors (generalized variance inflation factors).

Term	VIF
v_commune	1.12
region	1.54
wave	1.01
female	1.00
age10	1.00
ethnicity	1.16
education	1.00
urban	1.14
log_wt_c	1.09

M4 is not singular and converged without warnings.

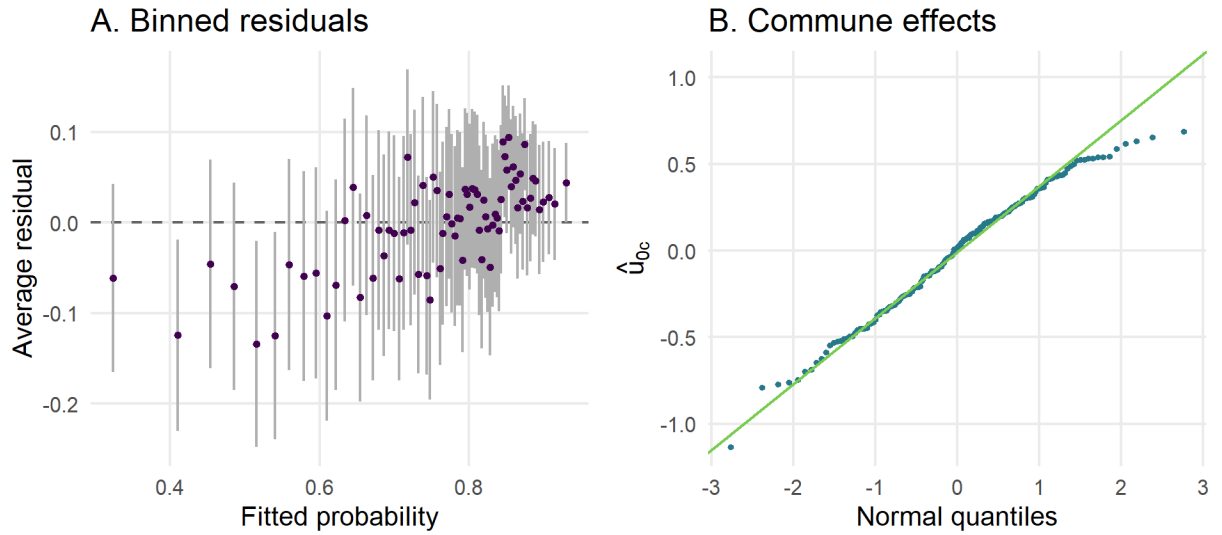
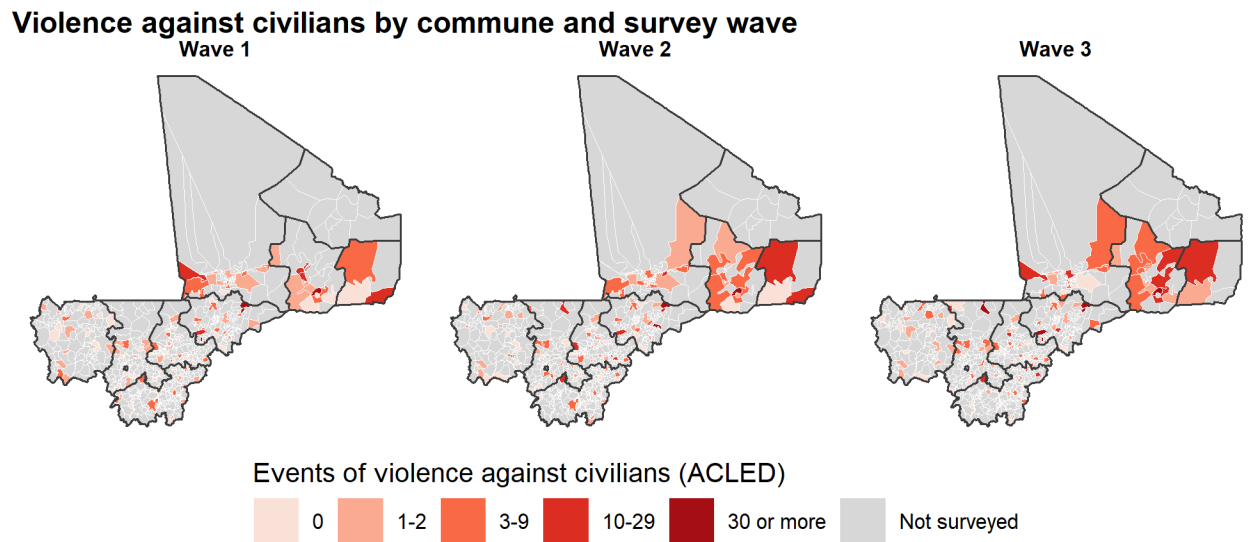


Figure 2: (A) M4 binned residuals: each point is the average residual in a bin of fitted values, with its 95% interval (gray); most intervals should cross zero (dashed line), with no pattern across fitted values. (B) M4 commune random effects against a normal distribution.

6 Maps

Gray communes were not surveyed in that wave: this is missing data, not zero.

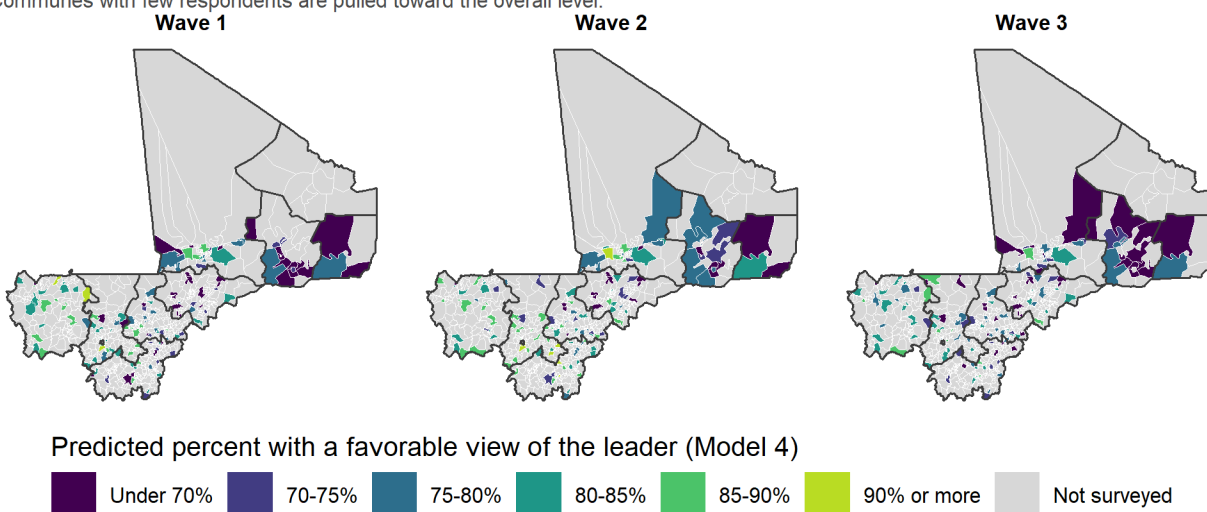


Boundaries: DNCT Mali (2021), via OCHA COD-AB, Humanitarian Data Exchange. Survey communes not placed on the map: 0.

Figure 3: Events of violence against civilians by commune and wave (surveyed communes only).

Predicted leader favorability by commune and survey wave

Model 4 predictions including each commune's and village's own effects, averaged over its respondents with survey weights. Communes with few respondents are pulled toward the overall level.



Boundaries: DNCT Mali (2021), via OCHA COD-AB, Humanitarian Data Exchange. Survey communes not placed on the map: 0.

Figure 4: Predicted percent favorable by commune and wave from M4, including each commune's and village's own random effects, averaged over its respondents with survey weights.

7 How to explain the result

Among people surveyed in the same region and survey round, those living in communes with more violence against civilians were less likely to view the leader favorably. If every commune had experienced no such events, about 82.3% would be expected to hold a favorable view; with 5 events, about 73.2%. This mostly compares different communes, so it shows an association, not proof that violence caused the change.

8 Limitations

- **Mostly between communes.** Violence varies mainly between communes, so the estimate mainly compares different communes rather than the same commune over time. Stable differences the covariates miss (remoteness, local history) could drive part of it; the M3-MW sensitivity model shows how much the within-commune change agrees.
- **Random slope.** With little within-commune variation, the random slope test has low power. Not rejecting it does not show that the effect is the same everywhere.
- **Unweighted multilevel models.** The multilevel models do not use the survey weights directly. M4 checks for informative sampling with the log weight, and the design-based check shows whether weighting changes the conclusion.
- **Region-wide shocks.** Region and wave dummies enter separately; M3b tests for region-by-wave shocks with a random intercept, which pools thin cells toward the main effects.

- **Violence measure.** Only surveyed communes have event counts. ACLED undercounts events in remote areas, which most likely weakens the association.
- **Coverage.** Insecure areas are harder to survey, so estimates describe the population that could be reached.
- **Map placement.** Communes are joined to the boundaries by name key; names shared by two communes in one region cannot be placed and are counted in each caption.

9 References

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